

ShanghaiTech University

EXAMINATION FOR
(Semester I : 2018/2019)

EE216 – RECONFIGURABLE COMPUTING

Jan 2019 – Time Allowed: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. This paper contains **EIGHT (8)** questions and comprises **FOUR (2)** printed pages.
2. All questions are compulsory. Answer **ALL** questions.
3. Total marks of all questions are **SEVENTY (70)**.
4. This is an **OPEN BOOK** examination.
5. Programmable calculators are not allowed.

Please answer the following questions based on the attached research paper.

- Q1. Write a title for this paper. (7 marks)
- Q2. Write an abstract for this paper. (7 marks)
- Q3. If the interconnect of the paper is partitioned into 3 sub-networks: $N_{9:1}$, $N_{7:1}$, and $N_{7:0}$, instead of its current 3 sub-networks: $N_{8:2}$, $N_{6:2}$, and $N_{6:1}$, analyze the pros and cons you may think of. (7 marks)
- Q4. Highlight the critical path of the DSP CLB directly in Figure 4b, and clearly draw how a signal is flowing along the critical path. (7 marks)
- Q5. Can we directly use traditional SRAM arrays in implementing the configuration memory of FPGAs? If not, why? (7 marks)
- Q6. The authors have specified that a custom place and route (PAR) tool is developed for their architecture. Why is a custom PAR necessary? What kind of changes need to be made to an existing tool to support their architecture? (14 marks)
- Q7. Draw a routing-resource graph for the architecture presented in Figure 2a considering only two levels of Switch matrices. Find a path for a connection from the first LUT to the last LUT in the figure. (14 marks)
- Q8. What are the consequences of having less routing resources in the overall architecture? (7 marks)

The research paper is attached from the next page.

END OF PAPER